

$$\mathcal{L}_{SM} = \mathcal{L}_{gauge} + \mathcal{L}_{fermions} + \mathcal{L}_{Yukawa} + \mathcal{L}_{Higgs}$$

$$\begin{aligned}
\mathcal{L}_{gauge} &= \mathcal{L}_{gluon} - \mathcal{L}_{weak} - \mathcal{L}_{supercharge} \\
&= -\frac{1}{4}(\partial_\mu G_\nu^a - \partial_\nu G_\mu^a)(\partial^\mu G^{a\nu} - \partial^\nu G^{a\mu}) \\
&\quad - \frac{g_s}{2} f^{abc}(\partial_\mu G_\nu^a - \partial_\nu G_\mu^a)G^{b\mu}G^{c\nu} \\
&\quad - \frac{g_s^2}{4} f^{abc} f^{ade} G_\mu^b G_\nu^c G^{d\mu} G^{e\nu} \\
&\quad - \frac{1}{4}(\partial_\mu W_\nu^i - \partial_\nu W_\mu^i)(\partial^\mu W^{i\nu} - \partial^\nu W^{i\mu}) \\
&\quad - \frac{g}{2} \epsilon^{ijk}(\partial_\mu W_\nu^i - \partial_\nu W_\mu^i)W^{j\mu}W^{k\nu} \\
&\quad - \frac{g^2}{4} \epsilon^{ijk} \epsilon^{ilm} W_\mu^j W_\nu^k W^{l\mu} W^{m\nu} \\
&\quad - \frac{1}{4}(\partial_\mu B_\nu - \partial_\nu B_\mu)(\partial^\mu B^\nu - \partial^\nu B^\mu)
\end{aligned}$$